Preparación PAES

Clase 06: Función Potencia

EruditoLab

Tu fórmula para el éxito

Eje Álgebra y Funciones
Función Potencia: Introducción

Una **función potencia** es toda función de la forma:

$$f(x) = a \cdot x^n$$

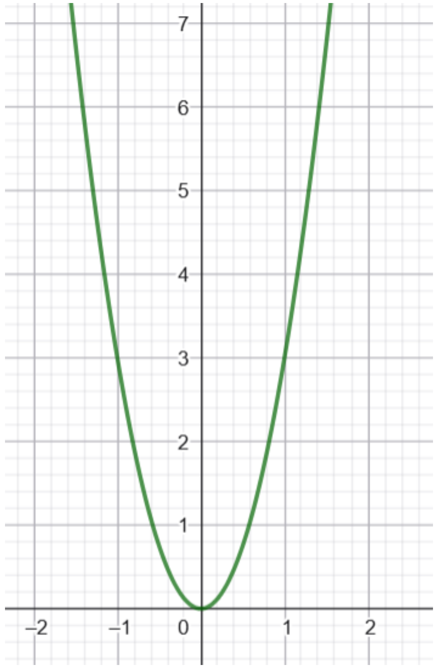
Donde:
a es una constante real $\neq 0$
n es un número real $\neq 0$

$f(x) = 3x^2$ $\qquad$ $f(x) = -5x^{-1}$ $\qquad$ $f(x) = 0{,}2x^{15}$

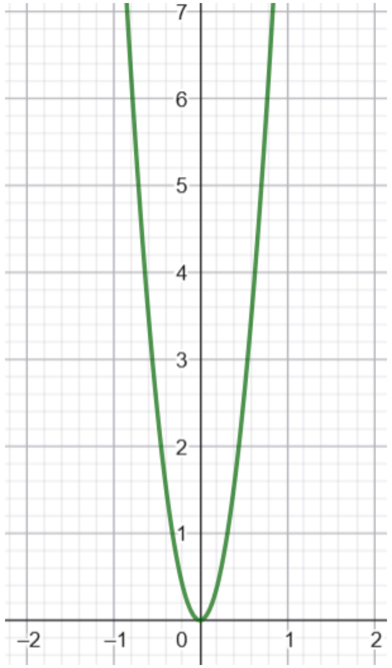
Existen distintos ejemplos de funciones potencia, dependiendo de los valores que tengan a y $n.$

Eje Álgebra y Funciones
Función Potencia

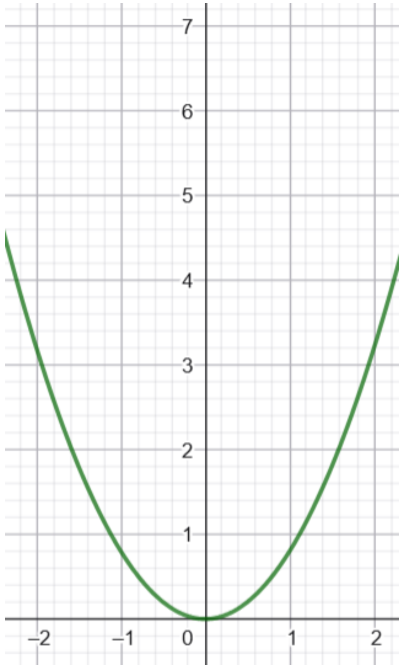
Caso 1: a > 0   n > 0 n es par



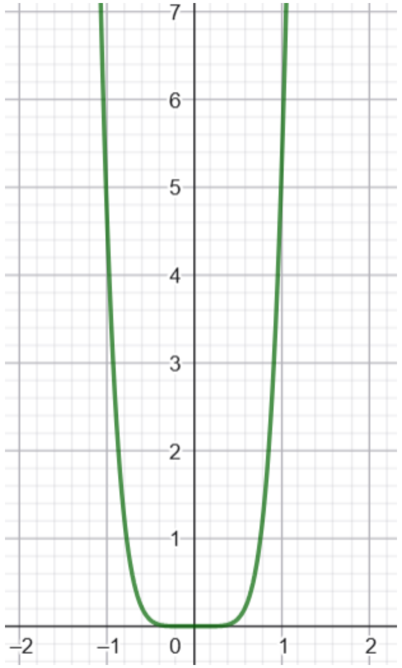
$f(x) = 3x^2$



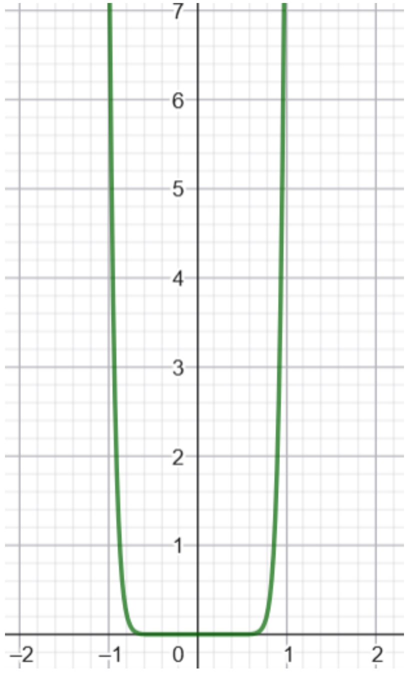
$f(x) = 10x^2$



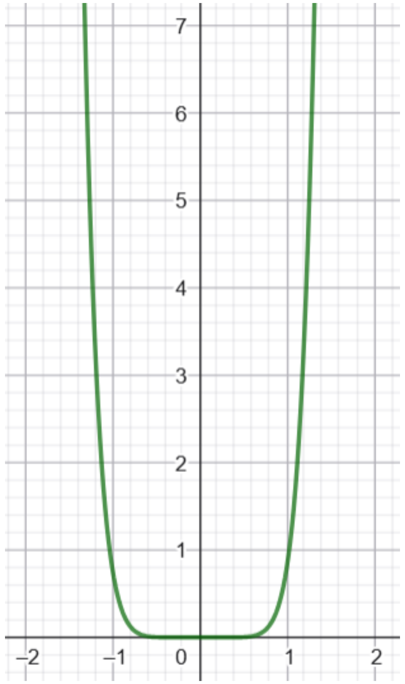
$f(x) = 0{,}8x^2$



$f(x) = 5x^6$



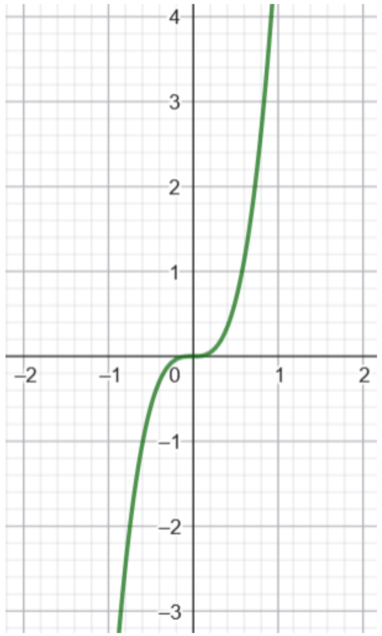
$f(x) = 10x^{16}$



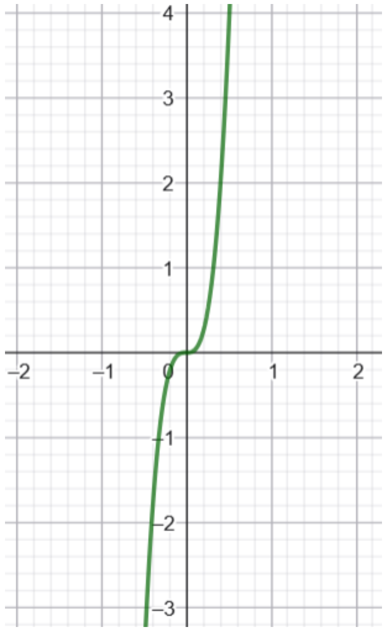
$f(x) = 0{,}8x^8$

Eje Álgebra y Funciones
Función Potencia

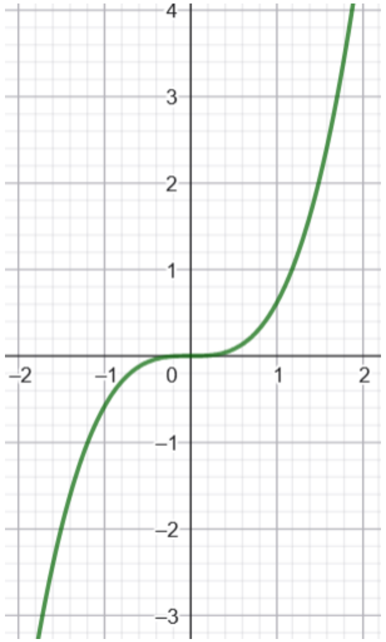
Caso 2: a > 0   n > 0 n es impar



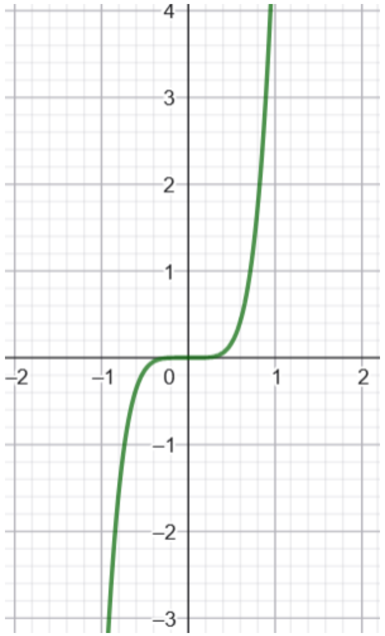
$f(x) = 5x^3$



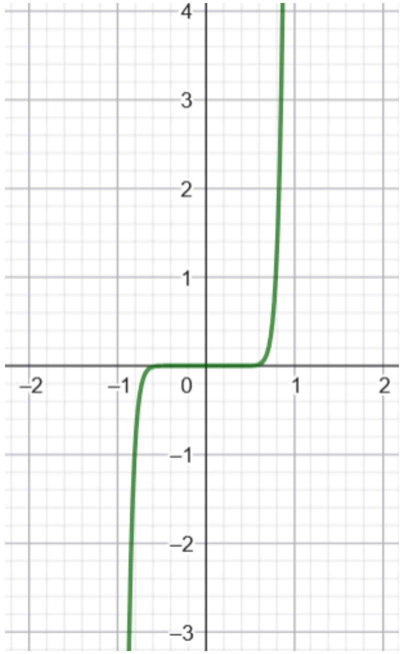
$f(x) = 30x^3$



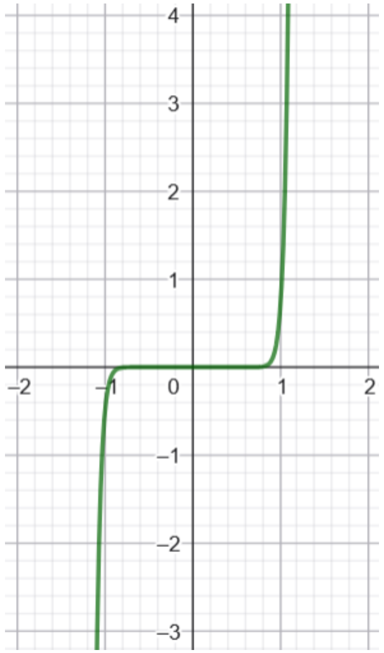
$f(x) = 0{,}6\ x^3$



$f(x) = 5x^5$



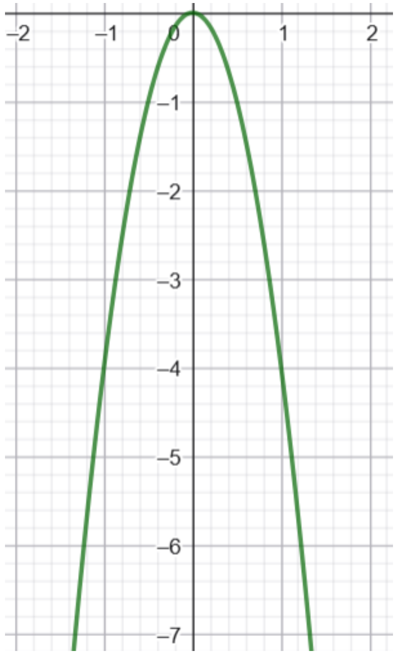
$f(x) = 30x^{15}$



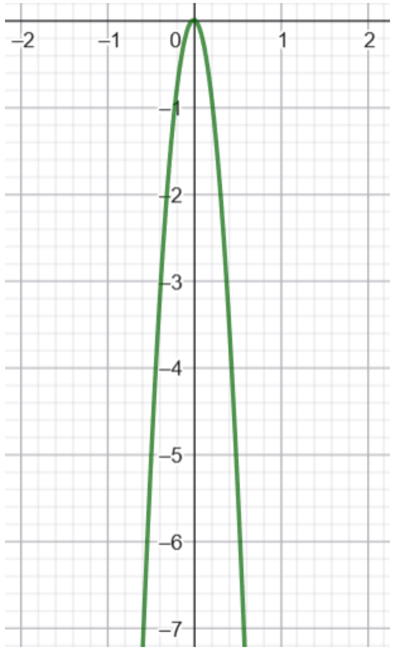
$f(x) = 0{,}6x^{21}$

Eje Álgebra y Funciones
Función Potencia

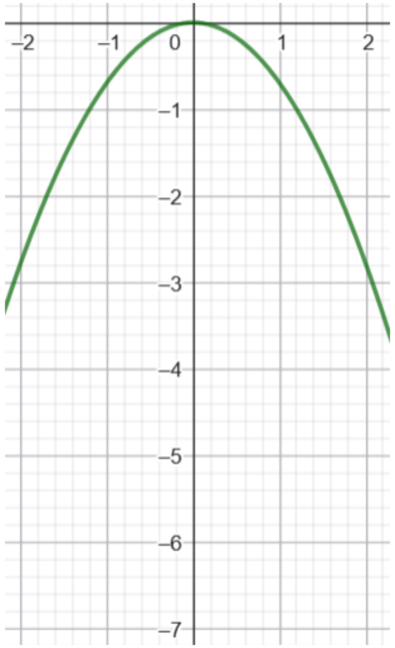
Caso 3: a < 0   n > 0 n es par



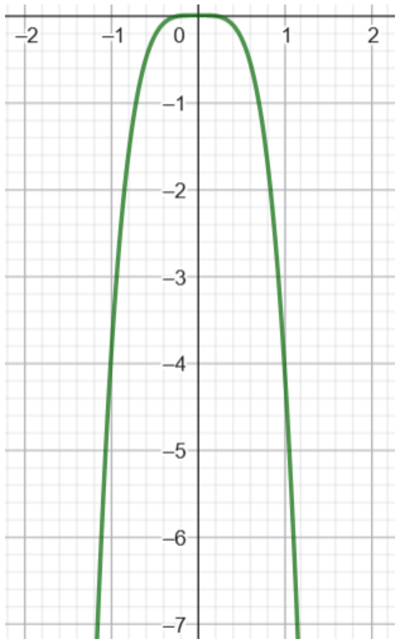
$f(x) = -4x^2$



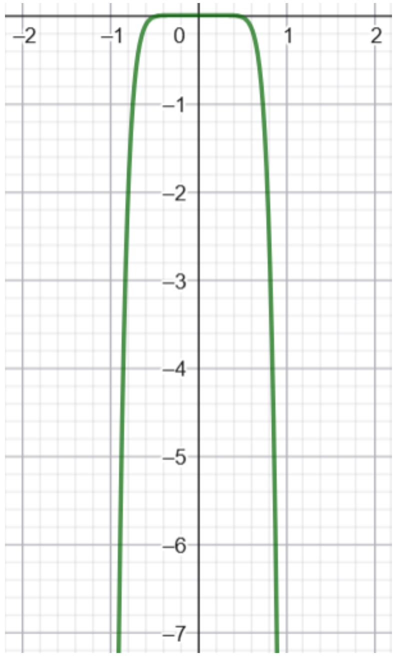
$f(x) = -21x^2$



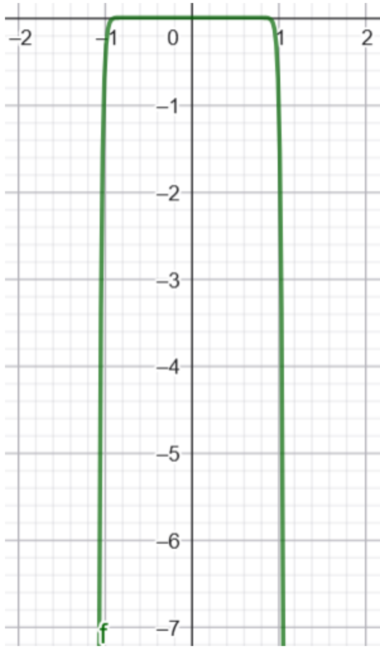
$f(x) = -0{,}7\ x^2$



$f(x) = -4x^4$



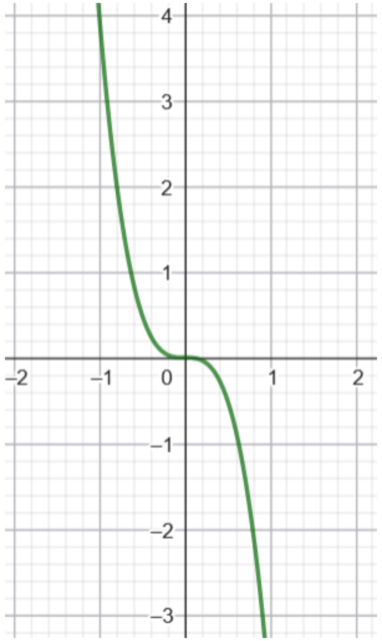
$f(x) = -21x^{10}$



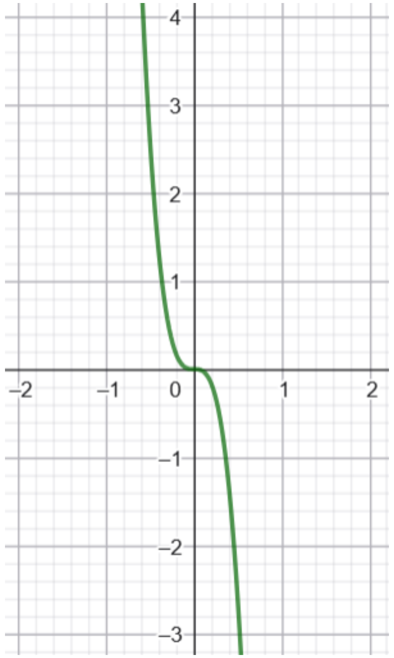
$f(x) = -0{,}7x^{40}$

Eje Álgebra y Funciones
Función Potencia

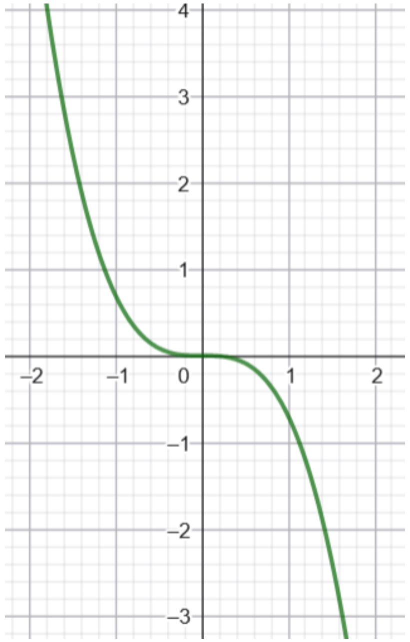
Caso 4: a < 0 n > 0 n es impar



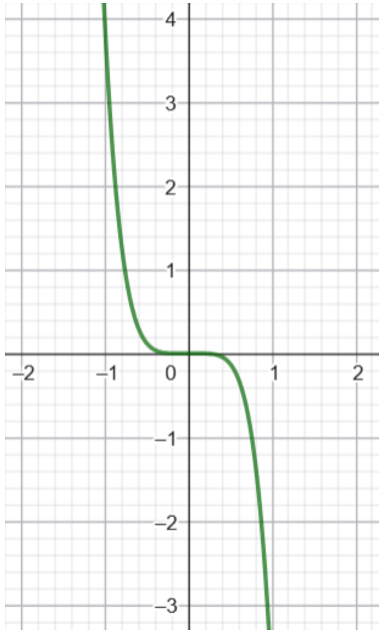
$f(x) = -4x^3$



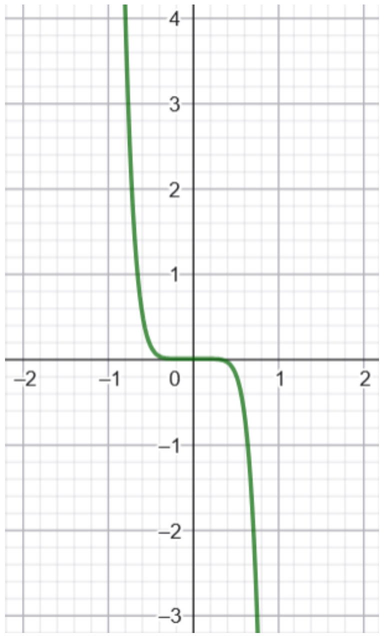
$f(x) = -21x^3$



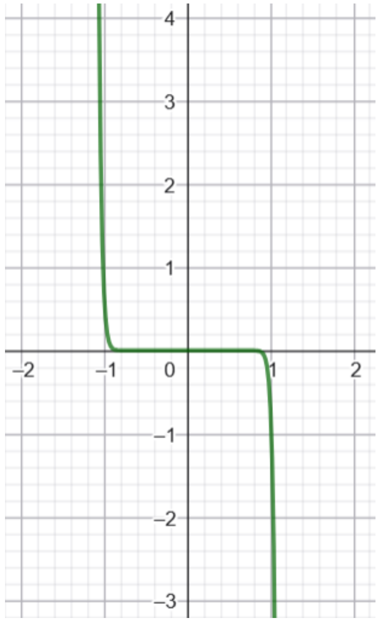
$f(x) = -0{,}7\ x^3$



$f(x) = -4x^5$



$f(x) = -21x^7$



$f(x) = -0{,}7x^{31}$

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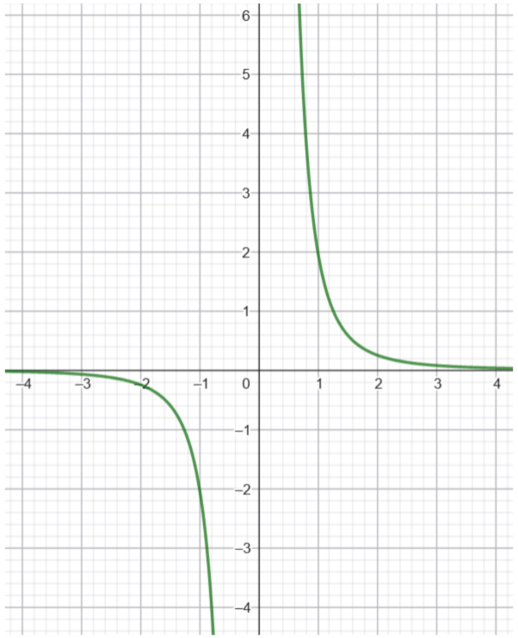
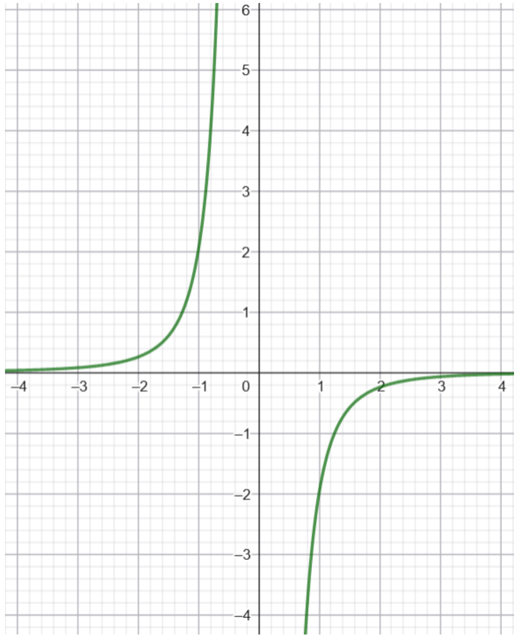
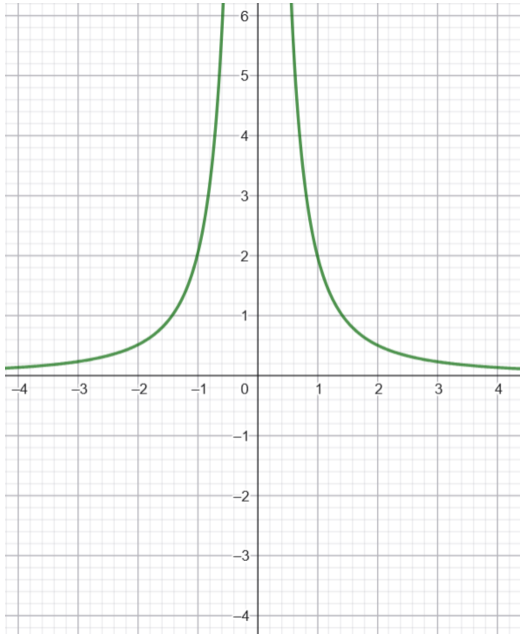
Caso 5: Casos indefinidos



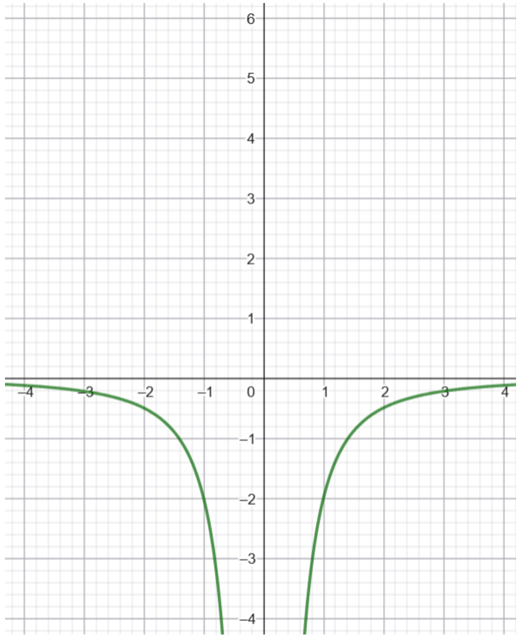
$f(x) = 2x^{-3}$



$f(x) = -2x^{-3}$



$f(x) = 2x^{-2}$



$f(x) = -2x^{-2}$